

Carrier Leads in Innovative & Sustainable Cooling



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Global commitment towards sustainability is guiding tech innovations across Industries. How has this approach played out in the cooling appliances industry?

Sustainability is no longer a peripheral commitment. It has become the central design imperative for HVAC industry. Buildings account for nearly 40% of global energy-related emissions, and HVAC systems contribute close to 15% of greenhouse gas emissions annually. That places the cooling industry at the very heart of the climate transition carrying both enormous responsibility and an innovation opportunity.

At Carrier, our response has evolved across three interconnected pillars: energy-efficient hardware, low global warming potential (low-GWP) refrigerants, and AI-powered digital platforms. And the results speak for themselves. Carrier's high-efficiency and lower-GWP products have helped customers avoid more than 490 million metric tons of GHG emissions since 2020, a figure that has grown significantly year on year. We have invested approximately \$1.6 billion in sustainable research and development since 2020 and have committed to investing over \$4 billion by 2030 in intelligent climate and energy solutions. India's own Cooling Action Plan projects HVAC energy demand to rise eightfold by 2038 making it imperative that every unit we bring to market is measurably more efficient than the one it replaces.

Carrier is a credible name in its segment. What were the differentiating innovations and market strategies implemented by you to keep the brand in the race?

Carrier invented modern air conditioning in 1902, and that legacy continues to shape how the company innovates, scales, and delivers value in evolving markets.

In India specifically, we have significantly expanded our Made-in-India portfolio domestically

manufacturing air-cooled and water-cooled chillers, VRF systems, packaged light commercial units, and hi-wall split systems to enhance accessibility, deepen our service network, and support channel partners across the country.

Our market strategy rests on lifecycle value, not just product sale. Through our Abound digital platform which now monitors over 1.1 billion sq. ft. of building space globally, we provide customers with predictive maintenance, energy audits, and carbon savings assurance. Carrier's sustainable products like Edge AI powered Chiller Plant Optimiser has helped customers realize up to 30% energy savings at the plantroom level, and we've seen large campuses standardize on the optimizer after measured step changes in kW/RT efficiency. Expanding on our local innovation commitment, we have also forged partnerships with institutions like Jamia Millia Islamia, Don Bosco to develop the next generation of skilled HVAC professionals, deepening our India ecosystem.

What are the different kinds of aircons Carrier has considered the diverse geo-climatic conditions that India has? Could you share some salient features and the affordability factor in a price-sensitive market?

India doesn't have uniform climate; it spans a wide spectrum. From the humid coastlines south to the dry heat of east, the temperate North, and humid-tropical tier-2 cities, each zone demands a tailored response. Carrier addresses this with a comprehensive portfolio of inverter-driven hi-wall splits for residential use, cassette units for commercial spaces, VRF systems for large multi-zone buildings, and chillers for industrial and district cooling applications. Our systems now incorporate adaptive airflow and humidity controls that automatically adjust to real-time outdoor conditions critical for India's seasonal extremes.

On affordability, our strategy is to democratise efficiency. Inverter technology ensures lower running costs, which over a product's lifetime often more than compensate for the upfront price differential. We are also expanding Cooling as a Service (CaaS), enabling customers to adopt high-efficiency cooling solutions without upfront capital investment. This model ensures predictable operating costs, continuous optimization, and measurable sustainability outcomes. Beyond product innovation, we are investing significantly in localized R&D, AI enabled controls, DCIM integration, and region specific solutions across Asia. Our manufacturing, engineering, and digital presence in India and APAC allows us to design region ready systems that address local operating conditions, regulatory frameworks, and sustainability targets.

AI and IoT has changed the concept of design, features, quality and durability. How is Carrier reorienting itself in this evolving hi-tech advancement?

AI has moved HVAC solutions from reactive to proactive. Where systems once waited for a fault to appear, they now anticipate, diagnose, and self-correct. Carrier's Abound platform exemplifies this by using AI and IoT technology. It has connected more than 45,000 chillers worldwide, identifying failure patterns before they occur and accelerating resolution. In one case, Abound delivered payback in under 12 months across 12 manufacturing and office buildings on a single campus. In 2024, we established a dedicated AI Centre of Excellence, a team of data scientists focused on our highest-impact use cases across operational efficiency and customer experience.

In India, our Abound IoT platform bridges HVAC with Building Management Systems, providing real-time insights that empower facility managers to take proactive decisions.

Carrier have also launched the IntelliSense Technology, an industry-first smart system that allows dealers to remotely monitor equipment health via a mobile app. And the results demonstrate less downtime, longer asset life, and lower total cost of ownership for the customer.

As a corporate veteran and of the cooling industry, according to you, what does it take to set brand benchmarks and the role of research, design, manufacturing and the most important people realize the objectives of the company?

At Carrier, we have over 6,000 engineers globally and operate six i3 Innovation Labs in the US, Poland, India, China, and Japan that function as incubators for disruptive technologies. More than 500 ideas have been generated through these labs, with active projects moving into business units.

Manufacturing plays an equally critical role. Nearly four decades of manufacturing in India have enabled us to build a deep and resilient ecosystem encompassing suppliers, channel partners, and system integrators. This localized strength allows us to respond faster to market needs, innovate closer to customers, and embed long-term reliability into our operations. Our Gurgaon manufacturing facility receiving IGBC Platinum certification reflects this commitment to sustainable and responsible manufacturing practices.

Beyond technology and scale, brand leadership also carries a responsibility towards communities and the environment. Our long standing partnership with the Indian Green Building Council, supporting schools through IGBC Green Your School program in their journey toward sustainability since 2007, reflects how we view our role as extending beyond products. In my experience, the formula is straightforward but demanding, research must be customer-obsessed, design must be climate-conscious, manufacturing must be lean and localised, and people must feel a genuine sense of purpose. India is a microcosm of that ambition.